A National Transformation Model: Value-Based Procurement for Wound Care

Commercial Proposal for NUPCO

Presented by:
SIGIL Scientific Enterprises

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CONFIDENTIAL

Executive Summary

Sigil Scientific Enterprises proposes a strategic partnership with NUPCO to implement a National Transformation Model for Value-Based Procurement in Wound Care, specifically focusing on Diabetic Foot Care.

This initiative is designed to shift the focus from traditional, volume-based purchasing of wound care products to an outcome-driven model that incentivizes healing, reduces complications, and optimizes the total cost of care.

The project will establish a comprehensive framework for Value-Based Healthcare (VBHC), beginning with a 12-month pilot focused on Diabetic Foot Ulcers (DFU), a condition with high prevalence and severe clinical and economic consequences in the Kingdom of Saudi Arabia.

By leveraging a proven, multidisciplinary approach that combines clinical expertise, data analytics, and strategic procurement, this project will deliver a scalable blueprint for managing chronic diseases, directly aligning with the healthcare transformation goals of Saudi Vision 2030.

Outcome-Driven Procurement

12-Month DFU Pilot Program

Multidisciplinary Execution Plan

Scalable National Blueprint

Aligned with Saudi Vision 2030

Strategic Alignment & Rationale

The Burden of Diabetic Foot Ulcers (DFU) in KSA

The Kingdom of Saudi Arabia faces a significant and growing burden from diabetes. The current procurement model, focused on unit cost, fails to address the **true cost of care**—the "cost of failure" associated with prolonged hospitalizations and severe complications.

18.3%
Adult Diabetes Prevalence

34%
Lifetime DFU Risk for Diabetics

31%
DFU Cases Lead to Amputation

42%
5-Year Mortality Rate Post-Amputation

Alignment with Saudi Vision 2030

Enhancing Healthcare Transformation: Supports the shift towards Accountable Care Organizations and integrated care models.

Promoting Fiscal Efficiency: Optimizes government spending by aligning procurement with NUPCO's mandate to pay for **value and outcomes**, not volume.

Improving Quality of Life: Reduces devastating complications like amputation and disability, directly improving patient well-being.

Scalable Blueprint: Establishes a successful VBP model for DFU that can be scaled to other chronic diseases across the Kingdom.

Patient Outcome Benchmarks

Measuring Value Through Clinical Results

Tier 1: Clinical

Patient-Level Outcomes

- Wound Healing Rate (12w)
- Time to Closure
- Amputation Rate
- Infection Rate

Tier 2: PROMs

Patient-Reported Outcomes

- Quality of Life (EQ-5D)
- Pain Reduction (VAS)
- Patient Satisfaction
- Functional Mobility

Tier 3: Economic

Cost-of-Care Impact

- Total Cost per Episode
- Avg. Length of Stay
- Supply Efficiency
- Cost per Healed Wound

Supplier Impact Assessment Framework

KPIs for Prequalification of VBP-Ready Suppliers

Suppliers entering the VBP programme must demonstrate measurable **impact on the pathology** and on the **healthcare providers**.

Impact on Pathology

- Wound Healing Efficacy
- Infection & Complication Reduction
- Patient Quality of Life
- Recurrence Prevention

Impact on Providers

- Clinical Training & Support
- Supply Chain & Logistics
- Cost Efficiency & Risk Sharing
- Innovation & Data Capability

Key Objectives & Deliverables

01 Establish Foundation

Develop a comprehensive VBP model for Diabetic Foot Care.

- Finalize KPI framework and baseline metrics.
- Deploy robust data infrastructure.
- Draft mutually agreed-upon VBP contract structure.

02 Align Stakeholders

Map and align all key stakeholders across the care pathway.

- Conduct alignment workshops with NUPCO and hospitals.
- Upskill clinical and administrative staff.
- Define new care pathways and protocols.

03 Execute VBP Pilot

Launch and manage a 12-month pilot in a live environment.

- Monitor clinical and economic KPIs in real-time.
- Demonstrate model effectiveness and value.
- Refine operational processes based on pilot data.

04 Create National Blueprint

Analyze outcomes and quantify total value created (ROI).

- Refine model into a scalable national VBP blueprint.
- Develop roadmap for scaling to other chronic diseases.
- Produce academic research and case studies.

Integrated Care Pathway & Scope

This project moves beyond simply procuring dressings and consumables; it involves procuring the entire patient outcome across a comprehensive, integrated care pathway.

Stage	Description	Key KPI
Screening & Prevention	Screening high-risk patients and providing patient education to prevent ulcer development.	Screening Coverage Rate
Early Detection	Ensuring rapid referral and initial assessment upon the first sign of an ulcer.	Time to Specialist Referral
Treatment	Providing advanced wound care, debridement, and infection control to promote healing.	Healing Rate (12 weeks)
Rehabilitation	Utilizing offloading, mobility support, and orthotics to prevent re-ulceration and disability.	Amputation Rate
Maintenance	Implementing long-term monitoring and preventative measures to avoid recurrence.	Recurrence Rate (1 year)

Project Roadmap: 12-Month Plan

Phase	Duration	Key Activities	Key Deliverables
Phase 1: Analysis & Design	**Months 1-3**	Stakeholder Mapping & Alignment Workshop; Current State Assessment; Patient Journey Mapping; KPI Framework & Baseline Definition; Draft Legal/Contract Framework	Stakeholder Map; Clinical Baseline Report; KPI Dictionary; Draft VBP Contract
Phase 2: Setup & Training	**Months 4-6**	Contract Structure Finalization & Signing; IT/Data Integration Setup; Clinical Staff Training on New Pathways; Supply Chain Logistics Setup & "Dry Run" Simulation	Signed VBP Contract; Live Data Dashboard; Staff Certification; Readiness Audit
Phase 3: Implementation	**Months 7-9**	Go-live of VBP Pilot; Real-time Monitoring of KPIs; Patient Enrollment & Tracking; Monthly Performance Reviews & Clinical Huddles	Monthly KPI Reports; Patient Registry; Issue Log & Resolution; Interim Financial Report
Phase 4: Evaluation & Scale	**Months 10-12**	Comprehensive Outcome & ROI Analysis; Total Cost of Care Calculation; Stakeholder Feedback Sessions; National Scaling Strategy Workshop	Final Impact Report; The National VBP Blueprint; Scaling Roadmap; Academic Research Paper; Case Study Video/Doc

Project Assumptions & Expectations

01 Phase 1: Analysis & Design

- NUPCO to provide initial list of internal contacts and access to historical data.
- NUPCO to nominate 8-12 clinical experts for Delphi surveys and provide existing contract templates.
- 2 online alignment meetings, 1 on-site visit (presencial), 1 online consensus workshop, 2 online review meetings.

02 Phase 2: Setup & Training

- NUPCO to coordinate final legal review and supplier negotiations.
- 2 online technical meetings, 2 training sessions (presencial), 1 on-site audit visit, 1 online review meeting.

03 Phase 3: Implementation

- Hospital to manage patient enrollment and consent protocols.
- 3 online monthly review meetings, weekly online check-ins (30 min), 1 presencial training session.

04 Phase 4: Evaluation & Scale

- NUPCO to organize venue for final presentation and provide target hospital info.
- NUPCO to facilitate Ministry of Health (MOH) input for the National Blueprint.
- Formal approval for scientific publication and case study video recording.
- 1 presencial presentation, 1 presencial workshop, 1 online planning session.
- 2 revision cycles for Final Impact Report. Journal fees excluded.

** For each deliverable, NUPCO will have 15 working days to provide feedback and request revisions. After this period, the deliverable will be considered approved.*

Commercial Model & Risk-Sharing

Four Critical Outcomes

Secure Data Infrastructure

Deployment of a NUPCO-specific, secure data processing instance with enterprise licensing and support.

VBHC Readiness & Capacity Building

Organizational culture assessment and upskilling of clinical teams in value-based principles.

VBHC Framework & RWE Strategy

Comprehensive study protocol, statistical planning, and a validated set of DFU outcome KPIs.

RWE Analysis & Publication Support

Execution of bespoke analysis algorithms and support for scientific publication of results.

Risk-Sharing Structure

- 80% Base Payment:
- Fixed payment tied to project milestones, covering foundational infrastructure, processes, and training.
- 20% Performance Bonus:
- Remuneration directly linked to the achievement of predefined clinical and economic outcomes (e.g., reduction in amputation rates).

Cost Structure: Phases 1 & 2

Phase 1: Analysis & Design (Months 1-3)

Subtotal: $235,000 USD / 881,250 SAR

Service	Price (USD)	Price (SAR)	Metric
Organizational Diagnosis & Capacity Building	$65,000	243,750	One-time for project
KPI Definition & Strategic Framework	$140,000	525,000	One-time for project
Legal/Contract Framework Draft	$30,000	112,500	One-time for project

Phase 2: Setup & Training (Months 4-6)

Subtotal: $425,000 USD / 1,593,750 SAR

Item	Price (USD)	Price (SAR)	Metric
Technology Infrastructure Setup	$30,000	112,500	One-time for project
Licenses & Infrastructure Maintenance	$185,000	693,750	1-Year
Algorithm & Dashboard Development	$195,000	731,250	One-time for project
Clinical Staff Training	$15,000	56,250	One-time for project

Cost Structure: Phases 3 & 4

Phase 3: Implementation (Months 7-9)

Total: $115,000 USD / 431,250 SAR

Service	Price (USD)	Price (SAR)	Metric
Consulting & Development Services	$50,000	187,500	One-time for project
Algorithm Execution & Monitoring Support	$65,000	243,750	One-time for project

Phase 4: Evaluation & Scale (Months 10-12)

Subtotal: $125,000 USD / 468,750 SAR

Item	Price (USD)	Price (SAR)	Metric
Technology Infrastructure Setup	$80,000	300,000	One-time for project
Contract Model Design & Finalization	$25,000	93,750	One-time for project
Scientific Publication Support	$20,000	75,000	Per Manuscript

Total Project Investment

Project Phase	Duration	Price (USD)	Subtotal (SAR)
Phase 1: Analysis & Design	Months 1-3	$235,000	881,250
Phase 2: Setup & Training	Months 4-6	$425,000	1,593,750
Phase 3: Implementation	Months 7-9	$115,000	431,250
Phase 4: Evaluation & Scale	Months 10-12	$125,000	468,750
Total Project Investment	**12 Months**	**$900,000**	**3,375,000**

Remarks

Travel, accommodation, and related logistics are estimated at 15% of the project fees. These expenses are not included in the fixed price above and will be invoiced monthly separately at cost (with supporting receipts).

Invoices related to expenses are due within 30 days of receipt.

Standard payment terms for the project fees listed above are Net 60 days from the invoice date.

VAT will be applied to invoices in accordance with KSA regulations.

The budget explicitly excludes academic journal submission/publication fees and any additional clinical training sessions beyond the scope specified in Workstream 2.

Payment Plan

Payment Milestone	%	Amount (USD)	Amount (SAR)
1. Project Kick-off	10%	$90,000	337,.500
2. Phase 1 Delivery: KPI Dictionary, Clinical Baseline Report & Draft VBP Contract	16%	$145,000	543,750
3. Phase 2 Delivery: Dashboard Go-Live, Staff Certification & Signed VBP Contract	47%	$425,000	1,593,750
4. Phase 3 Delivery: Patient Registry Launch, Monthly KPI Reports & Interim Financial Report	13%	$115,000	431,250
5. Final Delivery: Impact Report, National VBP Blueprint & Scaling Roadmap	14%	$125,000	468,750
Total Project Investment	**100%**	**$900,000**	**3,375,000**

Our Expertise: A Multidisciplinary Approach

To deliver this transformative project, **SIGIL Scientific Enterprises (SGL) acts as a strategic hub**, representing a consortium of world-class partners, each bringing a unique and essential set of capabilities to ensure success. This multidisciplinary team combines deep clinical knowledge, sophisticated economic modeling, and robust operational strategy to deliver a true end-to-end solution.

VBP Strategists

Experts in value-based contract design, risk-sharing models, and healthcare policy alignment.

Clinical Specialists

Specialists in diabetic foot care pathways, wound management protocols, and patient safety.

Data Architects

Professionals in health informatics, KPI dashboarding, and real-time outcome monitoring systems.

Powered by

sigil SCIENTIFIC ENTERPRISES | MTG RESEARCH & DEVELOPMENT LAB | medipal | Opvance | VISION FOR VALUE REACHING THE FULL POTENTIAL OF VBHC

Key Profile: Tiago Taveira-Gomes, MD, PhD

Tiago Taveira-Gomes

MD, MSc, PhD

Role in Project

Project Strategic Lead and Consortium Director

Position

Founder & CEO | SIGIL Scientific Enterprises / OPVANCE

Experience

Physician, data scientist, and serial entrepreneur with over 10 years of experience delivering large healthcare technology projects. Bridges the gap between clinical needs and technological execution.

Education

PhD in Medicine - Medical Education & Medical Informatics, University of Porto

MSc in Information Engineering - University of Porto

MD - University of Porto Faculty of Medicine

Post-graduation - Healthcare Management

Core Expertise

Real World Evidence & Implementation Science

Data Science & Machine Learning

AI in Healthcare

Preventive Medicine & Family Medicine

Key Profile: Filipe Costa, PhD

Filipe Costa

PhD

Role in Project

VBHC Lead

Key Achievements

Distinguished Merit Award from International Federation of Hospitals 2019

European prize for best VBHC Private group by EPHA

VBHC World Congress 2024 speaker

Position

CEO & Founder | Vision for Value; Professor at Nova SBE Executive Education

Experience

Over 25 years leading VBHC implementation projects at major institutions including MD Anderson Cancer Center, Hospital of Special Surgery, Santeon (Netherlands), and multiple NHS hospitals. Coordinator of Graduate Program in Health Management at Nova SBE.

Education

Post-Graduate - Value Based Healthcare, Harvard Business School

MBA

Master's in Management

Core Expertise

Value Based Healthcare Implementation

Healthcare Outcomes & TDABC

Business Plan Development & Strategic Planning

Performance Improvements & Project Management

Health Economics

Key Profile: Daniela Antunes, PhD

Daniela Antunes

PhD

Role in Project

Research & Strategy Lead

Key Achievements

Publications in Nature Communications, mBio, Microbiology Spectrum

Best Poster Presentation - 9th Advances Against Aspergillosis (2020)

Best Oral Presentation - 6th CESC Rising Stars in Mycology (2018)

Position

Senior Research and Strategy Analyst | MTG Research and Development Lab / SIGIL Scientific Enterprises / OPVANCE

Experience

Specializing in research strategy, grant management, and healthcare innovation. Strong background in immunology and infectious diseases. Previously R&D Consultant at FI Group, developing EU/National grant proposals.

Education

PhD in Immunology & Infectious Diseases - University of Minho / ICVS

Master of Biochemistry - University of Coimbra

Graduation in Biochemistry - University of Coimbra

Core Expertise

Research strategy development & translational research frameworks

EU and National grant proposal development

Scientific communication & high-impact publications

Strategic project management & stakeholder engagement

Key Profile: Ricardo Cruz-Correia, PhD

Ricardo Cruz-Correia

PhD

Role in Project

Technology Infrastructure Lead

Position

Co-founder & Associate Professor | Opvance / University of Porto

Experience

Over 20 years in health informatics. Recognized expert in Health Information Systems, Electronic Patient Records, and healthcare interoperability. PI in 26 research projects, supervised 50 graduate students, 348+ publications with 4,200+ citations.

Education

PhD in Human Biology - Faculty of Medicine, University of Porto

MSc in Computer Science - University of Porto

BSc in Computer Science - University of Porto

Core Expertise

Health Information Systems & Electronic Patient Records

Healthcare Interoperability (HL7, FHIR)

Medical Informatics & Telemedicine

Clinical Information Models & Standards

Data Science in Health

Key Profile: Carlos Sousa

Carlos Sousa

MsC

Role in Project

IT Governance Director

Certifications

COBIT® 5 Foundations; ECDL Examiner; Siebel Certified Consultant; Microsoft Certified Professional

Position

CEO & PMO Consultant | Opvance

Experience

Over 25 years as healthcare IT executive. Held Director-level positions at major Portuguese hospitals including Hospital Cruz Vermelha Portuguesa, Hospital Prof. Dr. Fernando Fonseca, and Hospital Distrital de Santarém. Expertise in digital health project management and governance.

Education

Diploma in Healthcare Management Program - Católica Lisbon School of Business and Economics

Post-Graduate in Health Information Systems - Universidade Católica Portuguesa

Degree in Computer Science - Universidade Autónoma de Lisboa

Core Expertise

Healthcare Information Systems Implementation

Digital Health PMO & Governance

IT Strategic Planning in Healthcare

Interoperability & Cybersecurity in Health

Key Profile: Marco Lima, MSc

Marco Lima

MsC

Role in Project

Data Engineering Specialist

Certifications

Oracle DBA 11g Certified Professional

Position

CTI and Systems Engineer | Opvance

Experience

Nearly 20 years at Centro Hospitalar Vila Nova de Gaia/Espinho. Proven ability to manage applications, solve complex problems, and drive process improvements through real-time data utilization. Specialized in SQL and data management within healthcare industry.

Education

Degree in Electrical and Computer Engineering - Automation, Control and Industrial Production Systems, Faculdade de Engenharia da Universidade do Porto

Core Expertise

SQL Development & Database Management

ETL Pipeline Development & Interoperability

Data Quality Monitoring

Healthcare Application Management

Clinical Information Systems Integration

Key Profile: Rafał Prymus

Rafał Prymus

MsC

Role in Project

Lead Frontend Architect

Position

Senior Frontend Developer | Sigil Scientific Enterprises

Experience

Senior Software Engineer with a strong background in building scalable SaaS applications and complex web interfaces. Previously at GlobalLogic and Ergonode, developed navigation/tracking software for global clients and custom Product Information Management tools. Expertise in streamlining customer workflows and ensuring system reliability across automotive and e-commerce sectors.

Education

Information Technology Electro-Electronic Technical School, Szczecin, Poland

Core Expertise

Modern Web Frameworks: Expert in Vue.js, Nuxt.js, React.js, and TypeScript

SaaS Development

Mobile & Native Builds

Schema-Driven Development: Ensuring consistent front-end/back-end integration

Key Profile: Mateusz Bułgajewski

Mateusz Bułgajewski

MsC

Role in Project

Lead Backend Architect

Position

Senior Backend Developer | Sigil Scientific Enterprises

Experience

Senior Backend Developer and Full Stack expert with over 10 years of experience. Proven track record in architecting modular backends, managing multi-stage CI/CD pipelines, and integrating complex payment and content management systems. Extensive background in PHP (Laravel), Python (FastAPI), and deploying scalable solutions on AWS for global clients.

Core Expertise

Modular design with FastAPI, JWT/API key authentication, and repository patterns.

Docker-based CI/CD pipelines deployed to AWS EC2 using GitHub Actions.

Seamless real-time collaboration using Laravel, Vue.js, and Inertia.js.

Integration of crypto/forex payment APIs and bespoke CMS development.

Key Profile: João Gancho Figueiredo, MD

João Gancho Figueiredo

MD

Role in Project

Clinical Strategy Consultant

Position

Senior Clinical Consultant | Sigil Scientific Enterprises

Experience

Medical Doctor with extensive experience bridging clinical practice, health management, and digital health technology. Previously Medical Director at Mediceus, where he led the development of patient-centric health data platforms, focusing on interoperability standards and clinical interfaces. Brings strong payer-side perspective from advisory roles at MDS Group and Médis, combining clinical expertise with strategic health plan management.

Education

Post-graduation in Health Management Nova School of Business and Economics

Integrated Master in Medicine Faculty of Medicine, University of Lisbon

Specialization in Occupational Medicine (Universidade Nova de Lisboa)

Core Expertise

Health Interoperability

Experience in health plan consultancy and insurance frameworks

Translating clinical requirements into functional technical specifications for engineering teams

Bridging the gap between medical stakeholders and technical development.

Immediate Next Steps

01

Proposal Review & Feedback

NUPCO to review the commercial proposal and provide feedback on the technical scope and financial structure.

02

Formal Agreement

Drafting and signing of the partnership agreement, including the risk-sharing contract and data processing addendum.

03

Project Kick-off

Official project launch with a stakeholder alignment workshop and the commencement of Phase 1 (Analysis & Design).

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Thank You

Ready to transform healthcare procurement in the Kingdom of Saudi Arabia.

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